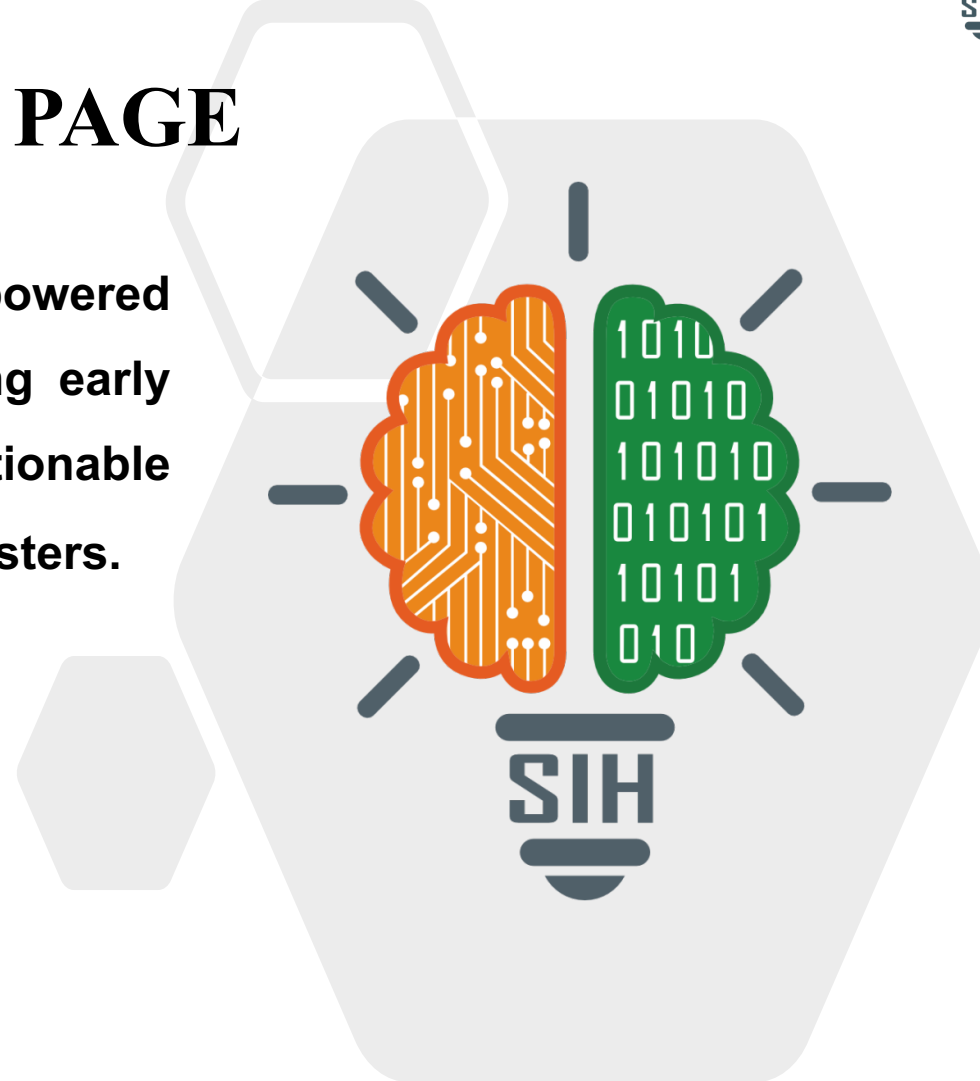


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TITLE PAGE

- Problem Statement ID – SIH26178
- Problem Statement Title- A resilient, AI-powered environmental monitoring network providing early detection, localized intelligence and actionable alerts for floods, forest fires and related disasters.
- Theme- Disaster Management
- PS Category- Hardware
- Team ID-
- Team Name- BHOOMI-NETRA





BHOOMI-NETRA

Predictive Environmental Intelligence Network



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IDEATION

- **The Problem:** Delayed disaster detection, complete telecom blackouts during extreme weather, and a lack of dynamic safe-routing for citizens.
- **Delayed Detection:** Cloud-dependent systems cause fatal delays and false alarms.
- **Network Blackouts:** Disasters destroy cell towers, severing critical warning alerts.
- **Static Routing:** Fixed maps fail when evacuation paths suddenly flood or catch fire.



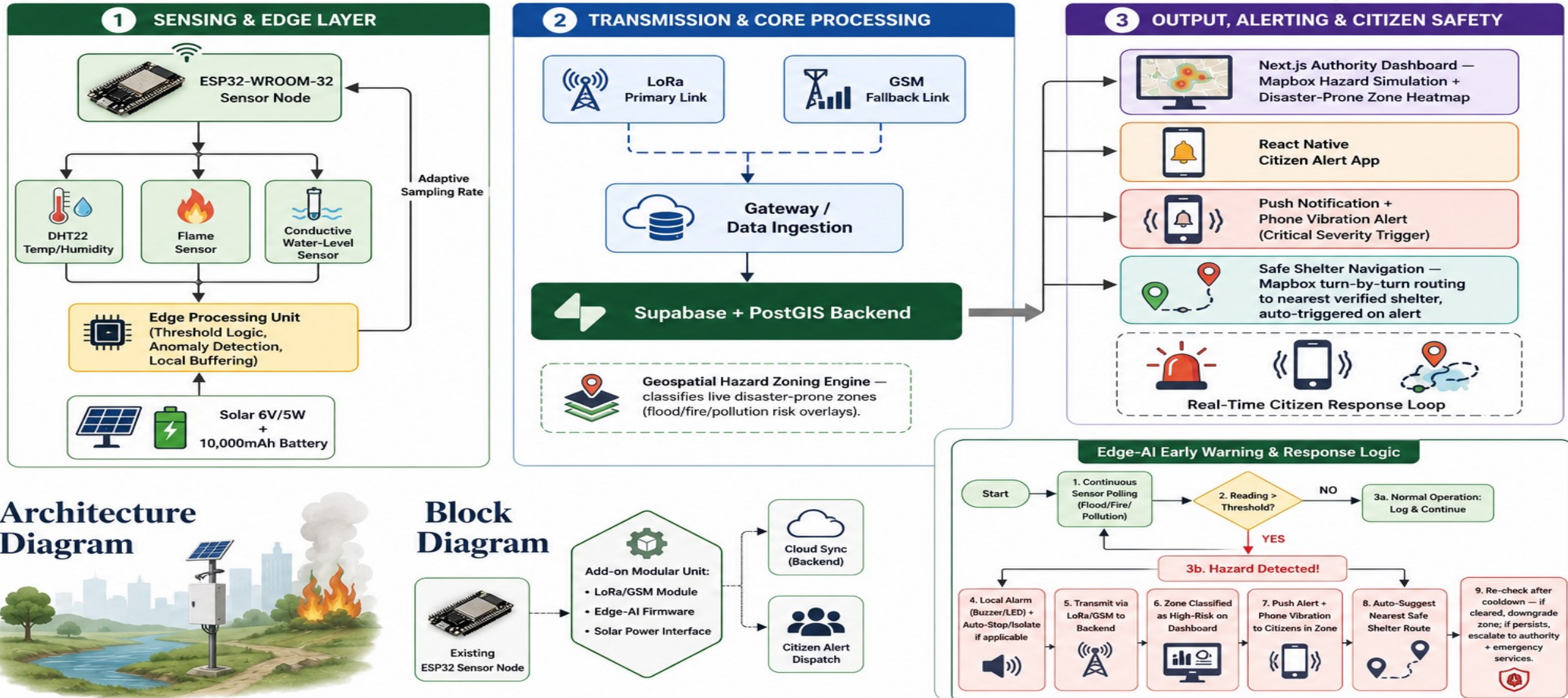
OUR SOLUTION (BHOOMI-NETRA)

- **A hybrid IoT sensor network and real-time mapping platform that detects floods and forest fires at the edge, simulating spread to issue actionable alerts.**
- **Edge-Validated IoT:** Autonomous nodes validate anomalies locally to eliminate false alarms.
- **Double-Tier Fallback:** System automatically shifts to a LoRa mesh if GSM towers fail.
- **Predictive Spread:** Uses DEM to forecast hazard paths and push dynamic, offline escape routes.



UNIQUENESS

- **Edge AI Validation:** Sensor anomalies are validated locally on the ESP32 (preventing false alarms) without waiting for cloud processing.
- **Adaptive Double-Radio Connectivity:** System automatically falls back through GSM and LoRa mesh, guaranteeing data transmission even if cellular towers go down.
- **Predictive Spread:** Integrates PostGIS Digital Elevation Models (DEM) to predict where floods or fires will spread next.





FEASIBILITY AND VIABILITY



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ANALYSIS OF FEASIBILITY

Technical Feasibility

- Built on standard ESP32, LoRa and GSM modules.
- IP65 enclosure ensures field durability.
- Open architecture — plug-and-play add-on boards.

Operational Feasibility

- Plug-and-play pole bracket → under-10-min deployment.
- 6 V / 5 W solar panel + 6600 mAh battery give 14-day zero-sun autonomy.



ANALYSIS OF VIABILITY

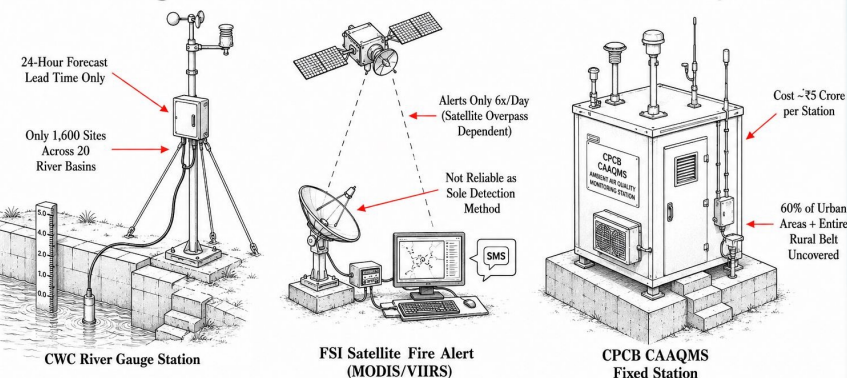
Economic Viability

- BOM ≈ ₹3,030 per node — far below conventional SCADA stations.
- Highly scalable; reuse existing towers and poles.

Environmental & Social Viability

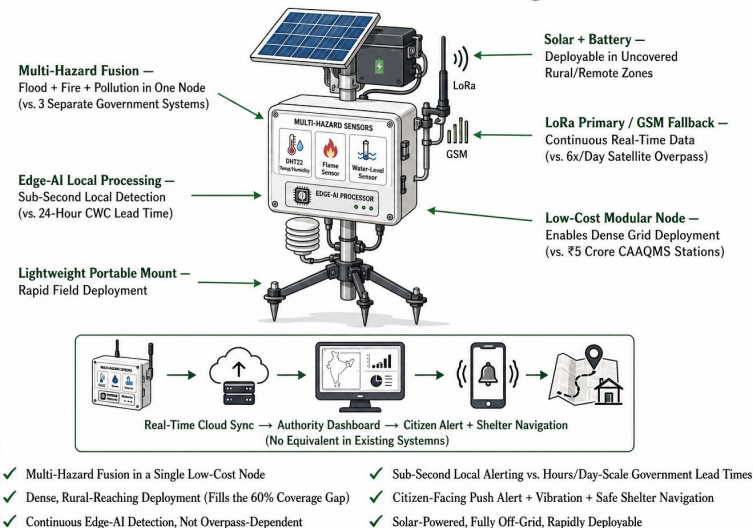
- Zero-grid dependence; non-invasive to forest ecology.
- Rural communities gain critical evacuation lead time.

Existing Solutions (CWC / FSI-FAST / CPCB-CAAQMS)



- ✗ Single-Hazard Systems (No Flood + Fire + Pollution Fusion)
- ✗ Sparse Ground Coverage — Rural & Remote Areas Excluded
- ✗ High Cost per Station Limits Scale
- ✗ Delayed/Non-Continuous Alerting (Satellite Overpass or Manual Reporting)
- ✗ No Citizen-Facing Local Alert or Shelter Guidance
- ✗ Fixed Infrastructure — No Rapid/Portable Deployment

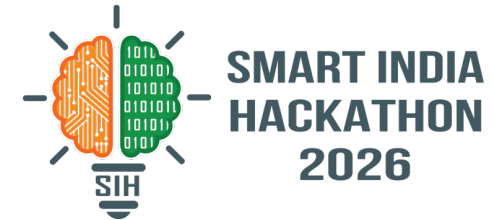
Our Solution — BHOOMI-NETRA Edge-AI Sensor Node



Feature	Existing Systems	BHOOMI-NETRA(Our)
Detection	Cloud-dependent	Edge AI local validation
Connectivity	Single-link	LoRa/GSM failover
Alerts	Generic sirens	Vibration + SMS alerts
Power	Grid-dependent	100% off-grid (Solar)



IMPACT AND BENEFITS



IMPACTS

For Disaster Authorities (Web Dashboard)

Predictive Command: PostGIS terrain heatmaps let SDRF pre-position boats and fire teams where hazards are spreading.

Credibility: Edge-validated alerts drastically reduce false positives and cut response times.



For Citizens (Mobile App)

Unignorable Alerts: Critical warnings trigger physical phone vibration and lock-screen overrides.

Safe Evacuation: Turn-by-turn offline mapping guides users away from danger zones to designated shelters (SMS fallback included).



For the Environment

Holistic Defense: Catching ignitions early prevents massive forest loss, protects rural agriculture, and shields urban areas from toxic smoke.

Non-invasive: Solar-powered nodes leave no permanent footprint on fragile ecosystems.



BENEFITS



Government / State Disaster Authorities

- Predictive command view for proactive SDRF / NDRF deployment.
- Reduction in false alarms through on-device anomaly validation.
- PostGIS streams accepted by state GIS portals.



Citizens & First Responders

- Unignorable mobile alerts (vibration + lock-screen override).
- Offline-safe evacuation routes to designated shelters.
- Inclusive SMS fallback for users on feature phones.

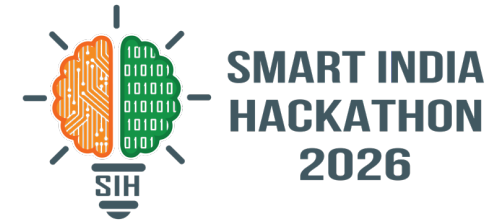


Environment & Ecology

- Early fire ignition containment reduces carbon loss.
- Solar-powered, zero-grid, non-invasive to forest ecology.



RESEARCH AND REFERENCES



FUTURE SCOPE

Satellite SAR Fusion

Combining ground-node data with satellite rainfall imagery for macro-scale flood & fire prediction.

TinyML Vision

Adding ultra-low-power, wake-on-event cameras for visual smoke / flame confirmation.

CAP 1.2 Integration

Upgrading to Common Alerting Protocol for unified government broadcasting across states.

LoRaWAN Expansion

Scaling beyond IN865 (865–867 MHz) for cross-border regional disaster monitoring pilots.



RESEARCH

Foundation & Compliance:

Architecture aligned with NDMA Guidelines on Disaster Risk Reduction; uses FSI historical wildfire hotspot data and regional flood-pattern records.

Hardware Standards:

LoRaWAN IN865 (865–867 MHz) protocol compliant for India; IP65 / IEC 60529 enclosure standards; BIS / WPC aligned for radio and solar components.

Edge AI References:

TinyML ESP32 inference benchmarks, ONNX-runtime-micro deployments, and PostGIS raster DEM flood-spread simulation literature.